Multi-objective Startup Fund

In this exercise, the Startup Fund problem is extended with an evaluation of each investments Sustainable Development Goal and the bi-objective version of the problem is sought, in different ways, over 4 steps.

Problem 1

- First, find the investment with maximal SDG

Parameters

- SDG_i: The SDG rating 1-10, where 10 is best, of the investments

  Then we first give the model for the first optimization:

Model

Objective:

- Maximal SDG rating:

$$\sum_i SDG_i \cdot x_i$$

Constraints:

- The budget has to be obeyed:

$$\sum_i CapitalRequirements_i \cdot x_i \leq Budget$$

Problem 2

- Keeping the maximal SDG value, maximize the profit

- The budget has to be obeyed:

$$\sum_i SDG_i \cdot x_i \geq SDG - MAX$$

Then the objective is changed to:

- Maximal profit:

$$\sum_i EstProfit_i \cdot x_i$$

```
#*************************************************************************
# MO Startup Fund 3, 1 assignment, "Mathematical Programming Modelling" (42112)
using JuMP
using HiGHS
using MultiObjectiveAlgorithms
#*************************************************************************


#*************************************************************************
# PARAMETERS
Investments = [1 2 3 4 5 6 7 8 9]
I=length(Investments)
#*************************************************************************


#*************************************************************************
# Data
EstProfit =     [ 29  35  24  52  53  41  43  68  28]
SDG =           [  8   6   8   3   4   5   3   2   7]

CapitalRequirements = [ 17  25  19  25  28  23  29  31  18]
Budget = 100
#*************************************************************************


#*************************************************************************
# Model
Investment = Model(HiGHS.Optimizer)

@variable(Investment, x[1:I], Bin) # 1 if investment is made in i

# Maximize SDG
@objective(Investment, Max,
           sum(SDG[i]*x[i] for i=1:I) )

# budget constraint
```

```julia
@constraint(Investment,
            sum(CapitalRequirements[i]*x[i] for i=1:I) <= Budget)

print(Investment)
#**************************************************************************


#**************************************************************************
# solve
optimize!(Investment)
#**************************************************************************


#**************************************************************************
# Report results
println("-------------------------------------");
if termination_status(Investment) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective SDG = $(sum(SDG[i]*value(x[i]) for i=1:I))")
    println("The resulting profit = $(sum(EstProfit[i]*value(x[i]) for i=1:I))")
    for i in 1:I
        println("Investment $(Investments[i]) $(round(Int8,value(x[i])))")
    end
else
  println("  No solution")
end
println("--------------------------------------");
#**************************************************************************
```

Comments to the Julia/JuMP program:

- Notice how it is relatively simple to perform lexiographic optimization: It is easy to add an extra constraint, and the right hand side is simply the first optimum found. When the second objective is added it overwrites the first objective.

Problem 2

Do exactly the same, but in the opposite order: First maximize profit, then insert a constraint forcing the profit to stay at max, update the objective function to maximize SDG.
Lexiographic opt: Profit-MAX: 191 Profit-MAX: 19

```julia
#**************************************************************************
# MO Startup Fund 3, 2 assignment, "Mathematical Programming Modelling" (42112)
```

```julia
using JuMP
using HiGHS
using MultiObjectiveAlgorithms
#*********************************************************************


#*********************************************************************
# PARAMETERS
Investments = [1 2 3 4 5 6 7 8 9]
I=length(Investments)
#*********************************************************************


#*********************************************************************
# Data
EstProfit =      [ 29  35  24  52  53  41  43  68  28]
SDG =            [  8   6   8   3   4   5   3   2   7]

CapitalRequirements = [ 17  25  19  25  28  23  29  31  18]
Budget = 100
#*********************************************************************


#*********************************************************************
# Model
Investment = Model(HiGHS.Optimizer)

@variable(Investment, x[1:I], Bin) # 1 if investment is made in i

# Maximize SDG
@objective(Investment, Max,
           sum(SDG[i]*x[i] for i=1:I) )

# budget constraint
@constraint(Investment,
            sum(CapitalRequirements[i]*x[i] for i=1:I) <= Budget)

print(Investment)
#*********************************************************************


#*********************************************************************
# solve
optimize!(Investment)

@constraint(Investment, sum(SDG[i]*x[i] for i=1:I) >= objective_value(Investment))
```

```julia
# maximize profit, under the constraint that SDG cannot be reduced
@objective(Investment, Max,
           sum(EstProfit[i]*x[i] for i=1:I) )

optimize!(Investment)

#************************************************************************


#************************************************************************
# Report results
println("-------------------------------------");
if termination_status(Investment) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective SDG = $(sum(SDG[i]*value(x[i]) for i=1:I))")
    println("The resulting profit = $(sum(EstProfit[i]*value(x[i]) for i=1:I))")
    for i in 1:I
        println("Investment $(Investments[i]) $(round(Int8,value(x[i])))")
    end
else
  println("  No solution")
end
println("--------------------------------------");
#************************************************************************
```

Problem 3

Now generate the opposite lexiographic optimal solution, i.e. Profit-SDG
Lexiographic opt: Profit-MAX: 191 Profit-MAX: 19

```julia
#************************************************************************
# MO Startup Fund 3 assignment, "Mathematical Programming Modelling" (42112)
using JuMP
using HiGHS
using MultiObjectiveAlgorithms
#************************************************************************


#************************************************************************
# PARAMETERS
Investments = [1 2 3 4 5 6 7 8 9]
I=length(Investments)
#************************************************************************
```

```julia
#*********************************************************************
# Data
EstProfit =      [ 29  35  24  52  53  41  43  68  28]
SDG =            [  8   6   8   3   4   5   3   2   7]

CapitalRequirements = [ 17  25  19  25  28  23  29  31  18]
Budget = 100
#*********************************************************************


#*********************************************************************
# Model
Investment = Model(HiGHS.Optimizer)

@variable(Investment, x[1:I], Bin) # 1 if investment is made in i

# Maximize SDG
@objective(Investment, Max,
           sum(EstProfit[i]*x[i] for i=1:I) )

# budget constraint
@constraint(Investment,
            sum(CapitalRequirements[i]*x[i] for i=1:I) <= Budget)

print(Investment)
#*********************************************************************


#*********************************************************************
# solve
optimize!(Investment)

# ensure that the found profit solution does not get worse,
# when optimizing regarding SDG
@constraint(Investment, sum(EstProfit[i]*x[i] for i=1:I) >= objective_value(Investment))

# Maximize profit
@objective(Investment, Max,
           sum(SDG[i]*x[i] for i=1:I) )

optimize!(Investment)
#*********************************************************************


#*********************************************************************
```

```
# Report results
println("--------------------------------------");
if termination_status(Investment) == MOI.OPTIMAL
    println("RESULTS:")
    println("objective SDG = $(objective_value(Investment))")
    println("The resulting profit = $(sum(EstProfit[i]*value(x[i]) for i=1:I))")
    for i in 1:I
        println("Investment $(Investments[i]) $(round(Int8,value(x[i])))")
    end
else
  println("  No solution")
end
println("--------------------------------------");
#************************************************************************
```

Problem 4

Finally, setup the full bi-objective model.

$$
\begin{array}{lrcl}
\texttt{Maximize.} & \sum_i EstProfit_i \cdot x_i & & \\
 & \sum_i SDG_i \cdot x_i & & \\
\texttt{Subject to :} & & & \\
 & \sum_i CapitalRequirements_i \cdot x_i & \leq & Budget \\
 & x_i \in \{0,1\} & &
\end{array}
$$

This is simply the model, with both objectives. This can be implemented more or less directly in Julia/JuMP:

```
#************************************************************************
# MO Startup Fund2 assignment, "Mathematical Programming Modelling" (42112)
using JuMP
using HiGHS
using MultiObjectiveAlgorithms
#************************************************************************


#************************************************************************
# PARAMETERS
Investments = [1 2 3 4 5 6 7 8 9]
I=length(Investments)
#************************************************************************


#************************************************************************
# Data
EstProfit =     [ 29  35  24  52  53  41  43  68  28]
```

```
SDG =           [ 8   6   8   3   4   5   3   2   7]

CapitalRequirements = [ 17  25  19  25  28  23  29  31  18]
Budget = 100
#*************************************************************************

#*************************************************************************
# Model
MO_Investment2 = Model()
@variable(MO_Investment2, x[1:I], Bin)

# setup two objectives: profit and SDG
@expression(MO_Investment2, profit_expr, sum(EstProfit[i] * x[i] for i=1:I))
@expression(MO_Investment2, SDG_expr, sum(SDG[i] * x[i] for i=1:I))
@objective(MO_Investment2, Max, [profit_expr, SDG_expr])

# budget constraint
@constraint(MO_Investment2, sum(CapitalRequirements[i] * x[i] for i=1:I) <= Budget)
#*************************************************************************


#*************************************************************************
# Solve
set_optimizer(MO_Investment2, () -> MultiObjectiveAlgorithms.Optimizer(HiGHS.Optimizer))
set_silent(MO_Investment2)

set_attribute(MO_Investment2, MultiObjectiveAlgorithms.Algorithm(), MultiObjectiveAlgorithms

optimize!(MO_Investment2)
#*************************************************************************


#*************************************************************************
# Report results
let

    if result_count(MO_Investment2)>=1
        println("No Pareto optimal points: ",result_count(MO_Investment2))
        for r in 1:result_count(MO_Investment2)
            z = objective_value(MO_Investment2; result = r)
            xx=round.(Int,value.([x[i] for i=1:I]; result = r))

            println("Profit: ",round.(Int,value.(z[1])), "    SDG: ",round.(Int,value.(z[2])
        end
    else
        println("No solutions found")
```

```
    end
end
#*************************************************************************
```